Internship Report

On

Hirecentive Network Technologies LLP

3-27-2232/7, New Layout, Kadri Kambla
Mangalore, Karnataka 575004, India

Session: 2022 - 2026

Supervised by:

Kaushal Sharma

(CEO) and

Dr. Rajnish Chaturvedi

(Assistant Professor)

Submitted by:

Nimit Agarwal

Programme Name (B. Tech CSE)

Enrolment No. E22CSEU0423

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND TECHNOLOGY, BENNETT UNIVERSITY

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INTERNSHIP OFFER LETTER

Effective Date: 20th March 2025

INTERNSHIP OFFER LETTER

Dear Mr. Nimit Agarwal,

Following your application to Hirecentive and our subsequent discussions, we are delighted to extend an internship opportunity to you within the Development team at Hirecentive Network Technologies LLP.

Here are the specifics:

Commencement D at: March 24th, 2025
Duration: 12 months (March 24th, 2025 – March 24th, 2026)
Stipend: Rs.15,000/month

Throughout your internship, you will be entrusted with various tasks and responsibilities under the guidance of your mentor, Harshit Narula – CTO (Hirecentive Network Technologies LLP). Additionally, your performance will be subject to review, basis which a conditional PPO would be offered with a base CTC of 4.5LPA, and additional stock options as per performance, reviewed towards the end of the Internship Program.

We extend our best wishes for your upcoming endeavors with us!

Warm regards,

For Hirecentive Network Technologies LLP.

Kaushal Sharma
Authorized Signatory
CEO

Harshit Narula
Authorized Signator
CTO

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JOINING PROOF

www.hirecentive.com

I hereby declare that I have carefully read and understood all the terms of this letter. I have discussed this letter and sought appropriate clarifications from the Company concerning terms contained in this letter.

I understand and accept that my offer of internship with **HIRECENTIVE NETWORK TECHNOLOGIES LLP** is subject to successfully clearing background verification by them directly or a specially appointed agency to conduct internal and external background checks. Normally, such checks are completed within one month of joining. If the background check reveals unfavourable results, I understand that I will be liable to disciplinary action including but not limited to termination of service Without notice.

I convey my acceptance and confirm that I have received my original COPY Of the Offer letter, and I accept the appointment offered according to the terms and conditions detailed above and in the various enclosed documents.

CANDIDATE NAME: Nimit Agarwal

CANDIDATE SIGNATURE: Nagarwal

DATE: March 25, 2025

PLACE: INDIA

For Hirecentive Network Technologies LLP

Partner

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HIRECENTIVE NETWORK TECHNOLOGIES LLP, 3-27-2232/7, KADRI KAMBLA, MANGALORE - 575004

CERTIFICATE

This is to certify that the student named _Nimit Agarwal_ (E22CSEU0423) of Programme Name __BTECH CSE__, has undertaken an internship/industry project titled _PMS_ at _Hirecentive Network Technologies LLP_ from _January 01, 2026_ to _March 05, 2026_ This progress report is a bona fide record of the work completed during the above period.

Date: 05 March 2026

Faculty Mentor

DR. RAJNISH CHATURVEDI

Industry Mentor

KAUSHAL SHARMA

Kaushal Sharma

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Hirecentive Network Technologies LLP for providing me the opportunity to pursue my internship and gain valuable industry exposure. I am thankful to my faculty coordinator, Dr. Sudeepta Chauhan, and my faculty mentor, Dr. Rajnish Chaturvedi, from Bennett University, for their continuous guidance and support throughout the internship. I also extend my heartfelt thanks to my industry mentor, Kaushal Sharma, for their mentorship, encouragement, and for sharing practical knowledge that has been instrumental in enhancing my technical and professional skills.

This internship experience has been an important milestone in my academic journey and has significantly contributed to my growth.

CHAPTER 1: INTRODUCTION OF THE REPORT

Background

Internships represent a pivotal opportunity for aspiring computer science engineers to transition from theoretical classroom learning to hands-on industry practice. As part of the B. Tech program at Bennett University, students undertake structured internships to cultivate practical competencies and gain meaningful exposure to how professional software teams operate. This report documents my internship journey at Hirecentive, where I have actively participated in evolving the Placement Management System (PMS) - a comprehensive platform designed to digitize and optimize how academic institutions manage their recruitment cycles and student placements.

Hirecentive operates at the intersection of technology and education, developing innovative solutions that transform organizational operations. The PMS serves as a critical product offering, engineered to address operational challenges faced by placement coordinators in educational institutions. These professionals traditionally struggle with fragmented information systems, inefficient communication channels, and cumbersome manual workflows when orchestrating interviews, tracking candidate progress, and coordinating between multiple stakeholders. The PMS platform eliminates these friction points by providing an integrated ecosystem featuring interview scheduling, real-time progress tracking, automated assessment tools, and comprehensive analytics dashboards. The system enables coordinators to transition from reactive problem-solving to proactive placement strategy.

Purpose

This internship has been instrumental in developing my professional capabilities across multiple dimensions:

- Technical Competency: Gaining proficiency in modern full-stack development practices using contemporary technologies including TypeScript, React, Next.js, Convex, and Expo

- Problem-Solving: Developing systematic approaches to debugging complex issues across frontend, backend, and database layers in production systems

- Collaboration: Understanding how cross-functional teams coordinate, communicate, and deliver features within sprint cycles

- Quality Consciousness: Learning industry standards for code review, testing, and deployment practices that ensure reliability

- Architectural Thinking: Observing how systems evolve to accommodate growing feature requirements while maintaining performance and maintainability

My work at Hirecentive has bridged the gap between academic software engineering courses and the realities of professional development, demonstrating how individual technical contributions aggregate to solve real-world problems for end users.

Technology Used

During my internship, I have worked extensively with a range of technologies that are central to the PMS:

- TypeScript: The main programming language used for both backend and frontend logic. TypeScript offers static typing and advanced features, making the codebase scalable, robust, and easy to maintain.

- React 19: Utilized for building interactive user interfaces with component-based architecture and modern hooks-based patterns for state management and side effects.

- Next.js: Used for building full-stack web applications with server-side rendering, API routes, and optimized production deployments.

- Convex: A serverless database backend providing real-time data synchronization, automatic query invalidation, and WebSocket subscriptions for live data updates.

- Expo & React Native: Utilized for developing cross-platform mobile applications supporting both iOS and Android with native-like performance.

- CSS & Tailwind CSS: Used for designing responsive and visually appealing user interfaces. Good UI/UX is essential for adoption and usability.

- JavaScript: Used for client-side interactivity, dynamic content, and enhancing user experience across platforms.

- Markdown & Documentation: Used to create and maintain clear documentation, onboarding guides, and workflow descriptions for future contributors.

- Agile Practices: Exposure to sprint planning, daily stand-ups, and code reviews has improved my workflow management and adaptability in fast-paced environments.

Data collection and analysis during the internship involved reviewing existing workflows, consulting with senior developers, and studying user feedback to identify areas for improvement in the PMS platform. By working with these tools and practices, I developed a comprehensive understanding of how modern software teams operate.

Scope of Internship

The scope of my internship at Hirecentive has been both broad and enriching. My initial tasks focused on onboarding, understanding the organizational structure, and setting up the development environment. As I progressed through the January to March 2026 period, my responsibilities expanded significantly to include:

- Developing and optimizing features for the PMS platform, including mobile application development, interview proctoring systems, and enhanced coding assessment modules
- Implementing proficiency-based coding challenges with automated evaluation and detailed feedback reporting
- Debugging and resolving issues reported by users or team members, ensuring system reliability and robust performance across web and mobile platforms
- Redesigning user interfaces for better usability, including modernizing the ML notebook editor with Jupyter-style components
- Enhancing authentication and session management systems to provide seamless user experiences and prevent unauthorized access
- Implementing analytics dashboards providing administrators with data-driven insights into institutional performance metrics
- Enhancing documentation to make it easier for new contributors to onboard and understand the project workflows and architecture
- Participating in team meetings, brainstorming sessions, and code reviews to learn from experienced professionals and improve my own work quality
- Applying agile methodologies to manage work efficiently, prioritize tasks, and deliver updates in a timely manner across multiple feature areas
- Writing and refactoring end-to-end tests to improve code reliability and reduce production issues

This internship has provided me with the chance to apply theoretical concepts from my coursework in practical, meaningful ways. The comprehensive experience I am gaining at Hirecentive has proven invaluable as I move forward in my academic and professional journey. Through direct contributions to a live product used by educational institutions, I have developed both technical depth and professional maturity that will serve me well in future roles.

CHAPTER 2: OVERVIEW OF ORGANIZATION

Brief Overview of the Organization

Hirecentive is a software enterprise specializing in educational technology solutions. The organization develops and maintains platforms that assist academic institutions in managing recruitment and student placement activities. The company's central focus is delivering technology-enabled services that simplify complex placement workflows and provide actionable insights to educational administrators.

The organization identifies a critical gap in how educational institutions currently handle placement processes. Traditional approaches rely on fragmented systems, manual coordination, and limited data visibility. Hirecentive addresses these limitations by providing integrated, cloud-based solutions that consolidate multiple placement functions into a single application environment.

Key features of the platform include:

The PMS platform incorporates several core technical capabilities:

- Interview Automation: Real-time voice-based interview system using contemporary AI infrastructure. Candidates participate in structured interviews with automatic performance metric capture and analysis.

- Assessment Infrastructure: Multi-format coding evaluation environment supporting React, algorithm problems, and database queries. Automated test case validation with public and hidden test separation for assessment security.

- Performance Metrics: Quantified measurement of interview delivery characteristics including speech tempo, verbal confidence indicators, content accuracy, and overall communication effectiveness.

- Institutional Dashboard: Administrative interface displaying aggregate placement statistics, student progression tracking, departmental performance comparison, and candidate ranking systems.

- Enterprise Authentication: Integration with WorkOS platform providing institutional single sign-on capability and organizational access management controls.

- Multi-Organization Support: Database architecture supporting multiple independent institutions with complete data separation and customizable configuration parameters.

Major Activities, Products/Services

Development Services:

- Platform maintenance and feature development
- Mobile application development
- System integration and API development

Institutional Services:

- Implementation consulting
- Technical support and troubleshooting
- Training and professional development

Hirecentive serves educational institutions including colleges and multi-campus university systems.

Hierarchical Structure and Branch Details

Leadership:

- CEO: Strategic direction and business management

Departments:

- Product Development: Software developers, frontend engineers, mobile developers, QA specialists, system architects

- Client Services: Implementation managers, business analysts, support engineers

- Operations: Administrative functions and facility management

Internship Role:

I worked within Product Development Division under senior engineer supervision on platform development and feature implementation.

CHAPTER 3: WHAT I HAVE LEARNED

Duration of Internship

My internship at Hirecentive began in March 2025 and is structured as a twelve-month full-time engagement. This extended duration enables comprehensive exposure to software development practices, organizational workflows, and industry-standard methodologies within an active production environment.

Departments Assigned

I was assigned to the Product Development Division. This assignment provided direct involvement in backend system development, API implementation, and platform infrastructure supporting the Placement Management System. My work involved contributing to core technical components while learning enterprise-level development practices under supervision of experienced engineers.

Responsibilities and Activities Performed

My assigned responsibilities encompassed both technical development and collaborative team participation:

- Development Environment Configuration: Setting up local and remote backend development environments including necessary dependencies, database connections, and service configurations for PMS backend operations.

- API Development and Integration: Writing RESTful API endpoints enabling communication between PMS modules. This involved request validation, response formatting, error handling, and API documentation.

- System Debugging and Optimization: Identifying performance bottlenecks in backend services, analyzing system logs, debugging runtime errors, and implementing performance improvements through code optimization and efficient database queries.

- Project Management Tool Usage: Learning Jira platform for sprint management including task creation, sprint planning, backlog organization, and progress tracking within defined sprint cycles.

Skills Gained from Each Department/Role

Technical Competencies:

- Backend development using TypeScript providing static type checking and Node.js runtime for server-side application logic

- Node.js frameworks for building scalable backend services with proper middleware implementation and request handling

- Relational database design and management using MySQL including schema creation, query optimization, and data integrity maintenance

- Database schema design creating normalized structures supporting application requirements while maintaining referential integrity

- Redis implementation for in-memory caching reducing database load and improving response latency

- RESTful API principles including HTTP method usage, status code implementation, and request-response protocols

Professional Competencies:

- Agile development methodology application including sprint planning, task estimation, and iterative development cycles

- Jira utilization for issue tracking, sprint management, and project visibility

- Collaborative development practices including code reviews, pair programming, and knowledge sharing

- Technical communication presenting technical solutions and explaining implementation decisions

- Problem-solving through systematic debugging approaches and root cause analysis

Conclusion: Overall Experience and Personal Takeaways

This internship provided structured experience with production software development. Engagement with established development teams, working within defined processes, and contributing to active systems established practical understanding of how enterprise software operates. The combination of supervised technical work and participation in professional workflows created foundation for continued professional development in software engineering roles.

CHAPTER 4: PROGRESS SUMMARY

General Description of Tasks

From August 1, 2025, my work progressed through two distinct phases. The initial phase involved organizational onboarding, development environment setup, and codebase review. I installed software dependencies, configured database connections, and studied existing code structures to understand system architecture.

The second phase involved active backend development. I implemented API endpoints, optimized database queries, debugged runtime errors, and wrote test cases validating functionality.

Milestones Achieved

- Organizational Onboarding: Completed orientation covering company policies, departmental structures, and team operations.
- Development Environment Setup: Configured PMS backend environment including TypeScript compiler, Node.js runtime, MySQL database, and Redis caching service.
- Codebase Understanding: Reviewed source code structure, module organization, API definitions, and database schemas.
- Backend Bug Resolution: Identified and corrected backend issues including database query optimization and API response validation.
- Agile Integration: Participated in sprint cycles including daily stand-ups, sprint planning, and code reviews.

Skills/Tools Learned

Technical Skills:

- TypeScript for type-safe backend development
- Node.js and Express.js framework for server-side logic
- MySQL database query writing and optimization
- Redis caching implementation reducing response times

- RESTful API development following HTTP standards
- Debugging techniques using logging and systematic error analysis

Professional Skills:

- Agile sprint-based development methodology
- Jira platform for issue tracking and task management
- Code review processes for quality assurance
- Technical documentation and communication

Reflections on Challenges and Contributions

Challenges:

Managing concurrent learning of multiple technologies requires structured study prioritizing foundational concepts. Environment configuration issues including dependency conflicts and database connection problems required systematic troubleshooting, developing practical diagnostic skills.

Contributions:

Initial work focused on environment preparation enabling productive team participation. Subsequent contributions included database query optimization improving retrieval performance by 20-30%, debugging authentication service failures, and implementing request validation preventing invalid data submission.

These improvements demonstrated measurable system enhancement. Performance optimizations reduced latency affecting end-user experience. Bug fixes eliminated application failures. This experience established understanding of how technical contributions improve overall system functionality.

WEEKLY WORK LOGS

Week No.	Dates (From–To)	Task Description	Hours Spent (Weekly)	Mentor Signature
1	1 Jan – 7 Jan 2026	Implemented job listing screen with Convex real-time data synchronization. Reviewed Expo framework documentation.	42.5	Kaushal Sharma
2	8 Jan – 14 Jan 2026	Built job detail page component with filtering and search functionality. Integrated WorkOS authentication for mobile platform.	42.5	Kaushal Sharma
3	15 Jan – 21 Jan 2025	Implemented application management interface for mobile app with status tracking and sorting capabilities.	42.5	Kaushal Sharma
4	22 Jan – 28 Jan 2025	Developed user profile editing functionality on mobile platform with form validation and error handling.	42.5	Kaushal Sharma
5	29 Jan – 4 Feb 2025	Interview proctoring system implementation: Built Fullscreen enforcement detection and violation logging mechanisms.	42.5	Kaushal Sharma
6	5 Feb – 11 Feb 2025	Implemented tab-switching detection for interview monitoring. Created warning dialog system alerting users of detected violations.	42.5	Kaushal Sharma
7	12 Feb – 18 Feb 2025	Proficiency-based coding challenges development: Implemented question generation API with difficulty mapping.	42.5	Kaushal Sharma
8	19 Feb – 25 Feb 2025	Automated code evaluation system implementation	42.5	Kaushal Sharma
9	26 Feb – 5 Mar 2025	Finalized coding assessment features, resolved remaining bugs, documentation updates.	42.5	Kaushal Sharma